STAT 527: Advanced Regression Analysis

Syllabus, Fall 2024

This version: October 18, 2024

An introduction to classical as well as modern regression methods, their motivation, justification, and implementation. Statistical properties of those methods, along with their advantages and limitations. Both theoretical and applied aspects of regression analysis will be discussed.

Lecture time and location:

10:00AM - 10:50AM MWF 2200 Sidney Lu Mech Engr Bldg

1 Contact Information

Instructor:

Jingbo Liu jingbol@illinois.edu
Office hours:
Fri 3:30pm-5pm Office 228 CAB

Teaching assistant:

Kaihong Zhang kaihong5@illinois.edu
Office hours:
Wed 3:30 - 5pm 2101 Everitt Laboratory. Wednesday from 3:30 PM to 5:00 PM in 2101 Everitt Laboratory.

2 Learning Management Systems

- UIUC Canvas: required; for distribution of course materials; submitting hw; forming project groups; send messages regarding course matters.

3 Prerequisites

STAT 410 or equivalent; STAT 510 (previous or concurrent enrollment); linear algebra; basic programming (preferably R, but other softwares are also fine).

4 Course Summary

Regression analysis aims at quantifying the relation between input variables (aka predictors or covariates) and the output variable (aka response). It is a fundamental topic of statistics, and is closely related to what is called supervised learning in computer science. The first half of the course revisits classical regression topics such as least squares, t-test, and model diagnostics, whereas the the second half gives a glimpse of more modern topics such as high-dimensional variable selection, robust regression and nonparametric regression. This is a PhD qualifying exam course, and for PhD students it is useful to get exposed to a variety of topics, both classical and modern, in order to smoothly read the literature when you conduct independent research in the future. Key objectives include:

- Design methods for model inference such as t-test and F-tests;
- Use R to run these tests and perform model diagnostics to test linearity, noise assumption or outliers;
- Practice literature reading and self-studying to delve deeper into a regression-related topic of your interest.

5 Course Materials and Textbooks

- Julian J. Faraway, "Linear Models with R", Chapman & Hall/CRC, 2015.
  Most closely connected to our lecture notes.
- Ashish Sen and Muni Srivastava, "Regression Analysis: Theory, Methods and Applications", Springer, 2011.
  Good reference on the theory side.
- Julian J. Faraway, "Extending the Linear Modol with R", Chapman & Hall/CRC, 2005.
- Trevor Hastie Robert Tibshirani Martin Wainwright, "Statistical Learning with Sparsity: The Lasso and Generalizations" `https://hastie.su.domains/StatLearnSparsity_files/SLS_corrected_1.4.16.pdf`
- Bradley Efron and Trevor Hastie, "Computer Age Statistical Inference Algorithms, Evidence, and Data Science," 2016.
  Contains good sections on FDR control.
- Ding, Peng. "Linear Model and Extensions." arXiv preprint arXiv:2401.00649 (2024).
  Good reference for general linear models (Chap 20-21)

We will provide comprehensive slides and lecture notes for the instruction materials. Many slides in the first half of the semester were borrowed from the previous instructor of the course (Naveen Narisetty), used with permission. Sharing course materials with any person who did not register the class, or uploading course materials to any website, is not allowed.

6 Weekly Schedule

1. 8-26,28,30: Linear regression, least squares estimator, projection interpretation.
2. 9-4,6: (skipped 9-2 labor day) Normal equations of least squares, standard error, Gauss Markov theorem, goodness of fit (R^2).
3. 9-9,11,13: Confidence interval for LS, standard error of prediction, hypothesis testing for a single coefficient, testing a subset of coefficients.
4. 9-16,18,20: Diagnostics in linear regression, residues, leverages, outliers, Cook's distance.
5. 9-23,25,27: Problems with the predictors, colinearity, non-i.i.d. or non-Gaussian noise.
6. 9-30,10-2,4: Transformation of predictors, variance stabilizing transformations, Box-Cox transformations.
7. 10-7,9,11: Take home midterm.
8. 10-14,16,18: Model selection, bias-variance tradeoffs, stepwise methods, criterion for selection (AIC, BIC, Mallows' C_p).
9. 10-21,23,25: LASSO, FDR control.
10. 10-28,30,11-1: Gaussian sequence model, selective inference.
11. 11-4,6,8: Debiased Lasso, quantile regression and robust regression.
12. 11-11,13,15: Analysis of variance (ANOVA), one-factor models, two-factor models,
13. 11-18,20,22: Nonparametric regression, bias-variance tradeoffs, linear smoother, KDE and local polynomials.
14. Fall break:
15. 12-2: Linear contextual bandit. 12-4,6,9,11 Project presentations.

The PhD qualifying exam tends to focus on the traditional topics covered in the class, excluding the software usage. Tentative plans for the last 2 weeks include project presentations by students or discussions of research topics.

7 Grading

- Homework (49%) Homework covers methods, theory, and basic programming in R. Solution files are due on Canvas, biweekly on Fridays midnights. Each hw 7 points (total 49), and occasionally there will be bonus questions for additional points. Below are the due dates:
  hw1: 9/13

hw2: 9/27
hw3: 10/11
hw4: 10/25
hw5: 11/8
hw6: 11/22
hw7: 12/6
Typesetting answers are highly recommended. Scanned versions are also ok, but should be highly readable. I will make a Canvas announcement when a homework is assigned. You should be able to configure your Canvas to receive emails about updates in the announcement section.

The solution file and your grades should be available after 7 days of the deadline, or slightly later.

Late homework policy: Please see the Policy section ahead.

- Midterm (26%) A take-home exam. The problems will be released on the Monday of the midterm week, and due on Friday of that week.

- Final project (25%) You can form groups of size ≥ 2 and ≤ 4, and choose one of the following two options.

  A Experimentation oriented: write codes to compare several algorithms on real data. Example:
  - compare different variable selection methods, the resulting type I and type II errors;
  - compare different choices of the M-estimator and run model diagnostics to compare the effects of these choices on the ability of dealing with outliers.
  - compare the high-dimensional (p/n fixed) versus the traditional (p fixed) theory predictions on real datasets, such as the asymptotic risk or the asymptotic normality phenomenon.

  B Literature oriented: read at least 3 papers on a common research topic related to regression analysis (see e.g. next section), and give a survey on the topic, including summary of results and future directions.

  Each group will complete *both* the following two:

  – write a 10 page report (due on Dec. 7)

  – make a presentation to the class in around 20 min.

  Grading is ultimately based on originality, clearity and depth of your work. But to make things concrete, I write a few general guidelines below:

  Grading guidelines for projects in category A are:

  (+10 points) problem setup and main ideas presented clearly and accessible to audience of the level of our class.

  (+12 points) Choose real datasets, and compare three statistical methodologies (within or beyond this course, such as from papers in the section below) for processing the data.

Implementation of each methodology should involve a reasonable level of sophistication. Gave insightful summary on advantages and limitations of the algorithms.

(+3 points) Make a publicly accessible demonstration of the project, e.g. codes and demo on Github.

Grading guidelines for projects in category B are (sum of the following points, with a cap of 25 points):

(+10 points) Choose a topic (e.g. "variable selection and multiple testing"), read about 3 papers on the topic, and present the main ideas clearly to audience of the level of our class.

(+5 points) discuss the relation between techniques in the papers and the materials covered in our course. What is really new in the papers?

(+5 points) show how the topic of the paper fits into the broader picture. Discuss connections to other topics in statistics/CS/math/engineering, and what are the future directions and open problems. When you are new to a field it is often impossible to go over all the literature or many details, but knowing there are significant open problems help you understand why it is useful to pursue that direction. You can list the open problems mentioned in these papers or based on your own thoughts.

This should be the most highlighted part of the project: demonstrate that you chose this direction because it is genuinely interesting and has a potential future impact, rather than because you found it easy/were asked by your advisor to do it.

(+5 points) includes new/simplified derivations of the results in the paper, or demonstrate the technique of the paper by solving a different, more illustrative example, which requires creative thinking.

(+10 points) contains original results and studies, nearly at the level of a new research paper (rather than some minor extensions of the paper being read).

8 Literature Guide for the Project

This is mostly for Category B final project mentioned above, although some papers also contain algorithms relevant for Category A. You are also welcome to find your own selection, e.g. by searching related papers using google scholar. However, it is recommended that you select papers from respectable sources, such as Annals of Statistics, Journal of the Royal Statistical Society, Probability Theory Related Fields, Annals of Probability, and Transactions on Information Theory, or top machine learning/IEEE/statistics conferences.

Variable selection, multiple testing, selective inference

1. Benjamini Y, Hochberg Y, "Controlling the false discovery rate: a practical and powerful approach to multiple testing". Journal of the Royal Statistical Society, 1995.
   The foundational paper in this field. Benjamini-Hochberg method is still widely today used due to its simplicity.

2. Benjamini Y, Yekutieli D, "The control of the false discovery rate in multiple testing under dependency". Annals of Statistics, 2001.
It extends the proof of the validity the Benjamini-Hochberg method from independent p-value case to the case of positive correlation. Moreover, it proposes a more conservative testing procedure (known as Benjamini-Yekutieli procedure `https://en.wikipedia.org/wiki/False_discovery_rate`) that works under completely general dependence structure.

3. N Johnson, "A study of the NIPS feature selection challenge," 2008. `https://hastie.su.domains/ElemStatLearn/comp.pdf`
This paper revisits the 2003 NIPS challenge, which was won by the Neal-Zhang group who used feature techniques to reduce the size of the network. In statistical learning methods such as regression tree and neural network, it is often beneficial to start with a large, overparameterized model and then perform model reduction, rather than starting from a small model.

4. R Berk, L Brown, A Buja, K Zhang, L Zhao, "Valid post-selection inference", *The Annals of Statistics*, 802–837, 2013.
This is one reference I found on "post-selection inference", i.e., how to perform inference after variable selection. You may search the key words for other related papers.

5. Taylor, Jonathan, and Robert J. Tibshirani. "Statistical learning and selective inference." Proceedings of the National Academy of Sciences 112.25 (2015): 7629-7634.

6. Lee, Jason D., et al. "Exact post-selection inference, with application to the lasso." ann. stats (2016): 907-927.

7. Rina Foygel Barber and Emmanuel Candès, "A Knockoff Filter for High-Dimensional Selective Inference", *Ann. Statist.*, 47 (2019).
A recent popular variable selection method; see more information on `https://web.stanford.edu/group/candes/knockoffs/`

8. Zrnic, Tijana, and Michael I. Jordan. "Post-selection inference via algorithmic stability." *Ann. Statist.*, arXiv preprint arXiv:2011.09462 (2020).

9. Spector, Asher, and William Fithian. "Asymptotically Optimal Knockoff Statistics via the Masked Likelihood Ratio." arXiv preprint arXiv:2212.08766 (2022).

High-dimensional analysis

1. Roman Vershynin, "High-Dimensional Probability An Introduction with Applications in Data Science," Cambridge University Press, 2018. `https://www.math.uci.edu/~rvershyn/papers/HDP-book/HDP-book.html`
Textbook on fundamental tools for high-dimensional analysis.

2. Martin J. Wainwright, "Sharp Thresholds for High-Dimensional and Noisy Sparsity Recovery Using ℓ_1-Constrained Quadratic Programming (Lasso)", IEEE Trans. Inf. Theory, 2009.

Typical high-dimensional statistics style work using concentration inequality bounds. You may find textbooks of the author on high-dimensional statistics or inference under sparsity.

3. A. Javanmard and A. Montanari. "Hypothesis testing in high dimensional regression under the Gaussian random design model: Asymptotic theory". *IEEE Transactions on Information Theory*, 60(10):6522-6554, 2014.
   Introduced ideas from statistical physics to high-dimensional statistics problems. Giorgio Parisi (who invented the replica method used here) won the Nobel Prize in 2021.

4. Pragya Sura, Emmanuel J. Candes, "A modern maximum-likelihood theory for high-dimensional logistic regression," PNAS 2018.
   The p/n = fixed framework (like the previous paper) can be extended to the logistic regression case.

5. PL Bartlett, PM Long, G Lugosi, A Tsigler, "Benign overfitting in linear regression," PNAS, 2019.
   A recent concept that motivated a lot of research activity. The goal is to explain the mysterious phenomenon that learning algorithms can generalize well even when the capacity is much larger than the traditional theory.

Adaptive designs, bandits

1. Lai, Tze Leung and Ching Zong Wei, "Least squares estimates in stochastic regression models with applications to identification and control of dynamic systems" *Annals of Statistics*, 10(1) 154-166, 1982.
   Earliest paper establishing consistency of regression under adaptive designs, and introduced fundamental tools such as the "elliptical potential lemma" widely used in analysis of online learning problems

2. T. L. Lai and Herbert Robbins, Asymptotically efficient adaptive allocation rules, 1985. (proves lower bound for any multi-arm bandit algorithm for a particular problem)

3. Sebastien Bubeck and Nicolo Cesa-Bianchi "Regret Analysis of Stochastic and Nonstochastic Multi-armed Bandit Problems," Foundations and Trends in Machine Learning, 2012.

4. Vianney Perchet, Philippe Rigollet, "The multi-armed bandit problem with covariates," Ann. Statistics, 2013.
   Regression method for incorporating side information in bandit problems.

5. V Perchet, P Rigollet, S Chassang, E Snowberg , "Batched bandit problems," Ann. Statistics, 2016.

Robust/quantile regression/M-estimator

1. Roger Koenker, "Galton, Edgeworth, Frisch, and prospects for quantile regression in economics" (August 1998). UIUC.edu. Retrieved August 22, 2018.
   Classic paper popularizing the quantile regression idea.

2. Portnoy, S. L. "Censored Regression Quantiles". *Journal of the American Statistical Association*, 98 (464): 1001–1012, (2003).

3. Yunwen Yang and Xuming He "Bayesian empirical likelihood for quantile regression", *Annals of Statistics*, (2010).

4. Kozumi, H., Kobayashi, G. "Gibbs sampling methods for Bayesian quantile regression", (2011).

5. Derek Bean, Peter J. Bickel, Noureddine El Karoui, and Bin Yu, "Optimal M-estimation in high-dimensional regression," PNAS, 2013. (This is also a mathematically sophisticated high-dimensional statistics paper. The PNAS paper does not contain proofs; for details of the proof see the PTRF version of the paper `https://link.springer.com/article/10.1007/s00440-016-0754-9`)

Nonparametric regression

1. A. Tsybakov, "Introduction to Nonparametric Estimation," 2008.

2. B Bilodeau, DJ Foster, DM Roy, "Minimax rates for conditional density estimation via empirical entropy," Annals of Statistics, 2023.
   Learning with side information is an important question; this paper discussed why conditional distribution estimation is tricky.

3. Tengyuan Liang, Alexander Rakhlin, "Just interpolate: Kernel 'Ridgeless' regression can generalize," Annals of Statistics, 2020.

Transfer learning and fine tuning

1. Craig, Erin, Mert Pilanci, Thomas Le Menestrel, Balasubramanian Narasimhan, Manuel Rivas, Roozbeh Dehghannasiri, Julia Salzman, Jonathan Taylor, and Robert Tibshirani. "Pretraining and the Lasso." arXiv preprint arXiv:2401.12911 (2024).
   Can you first pretrain a large model (e.g. CIFAR dataset with all kinds of animals), and then fine-tune it for the task of classifying cats and dogs only? I heard about it from Tibshirani's plenary talk at JSM 2024.

2. Li, Sai, T. Tony Cai, and Hongzhe Li. "Transfer learning for high-dimensional linear regression: Prediction, estimation and minimax optimality." Journal of the Royal Statistical Society Series B: Statistical Methodology 84.1 (2022): 149-173.

3. Cai, T. Tony, and Hongming Pu. "Transfer learning for nonparametric regression: Non-asymptotic minimax analysis and adaptive procedure." arXiv preprint arXiv:2401.12272 (2024).

8.1 Grading Scale

We will use a +/- scale:

- A+: [97, 100]
- A: [93, 97)
- A-: [90, 93)
- B+: [87, 90)
- B: [83, 87)
- B-: [80, 83)
- C+: [77, 80)
- C: [73, 77)
- C-: [70, 73)
- D+: [67, 70)
- D: [63, 67)
- D-: [60, 63)
- F: [0, 60)

9 Campus Policies

9.1 Academic Integrity

It is the responsibility of each student to refrain from infractions of academic integrity, from conduct that may lead to suspicion of such infractions, and from conduct that aids others in such infractions. In this course you are expected to complete the homework on your own. You may collaborate with your team on the project, and it is expected that everyone contributes to at least one essential part of the finished work (such as code writing, report writing, or presentation).

9.2 Disability Statement

"To obtain disability-related academic adjustments and/or auxiliary aids, students with disabilities must contact the course instructor and the Disability Resources and Educational Services (DRES) as soon as possible. To contact DRES, you may visit 1207 S. Oak St., Champaign, call 333-1970, e-mail `disability@illinois.edu` or go to the DRES website. If you are concerned you have a disability-related condition that is impacting your academic progress, you can talk with someone at the Counseling Center, McKinley Mental Health, or DRES about how to see a provider in order to obtain a diagnosis or get your questions answered."

9.3 Diversity and Equity

We support diversity, equity, and inclusion so that students from all backgrounds and abilities can succeed.
`https://las.illinois.edu/about/diversity/statement`

10 Course Policy & Tips

- Late homework:

  $\leq$1 day late: 10% reduction in grade.
  $\leq$3 days late: 20% reduction in grade.
  $\leq$7 days late: 30% reduction in grade.
  More than 7 days late: 0 grade because the TA will upload the solution by that time

  Exceptions: Extensions may be granted for documented emergencies or pre-approved reasons.

  Communication: Students must inform the instructor and the TA as soon as possible if they anticipate late submission. If TA has already graded a submitted version of the homework, we will not grade another updated version submitted after the deadline.

- Absence:
  Students whose absences meet the criteria may contact the Student Assistance Center in the Office of the Dean of Students to request an absence letter. For an excused absence due to illness, please bring a doctor's note.
  `https://studentcode.illinois.edu/article1/part5/1-501/`

- How to be successful?

  - Keep the objectives in mind. There are numerous references you can find on machine learning, with many details and variations on the algorithms, which can be overwhelming. It's useful to keep in mind the learning objectives while you read. Likewise, when you present or write about the project, please be very clear about a few key points you want to convey.
  - Be prepared to try. The best way to learn programming is by writing codes and trying things out on one's own. Similarly, learning the mathematical parts of the course can be made easy if we try to think about some basic examples and carry out calculations, rather than just memorizing the course materials.
  - Don't be shy to ask. We all come from different background, and there is no need to be afraid of saying you do not agree or requesting clarifications. Life favors the bold.